

Takashi Hikihara

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born August 9, 1958

Kyoto University Executive Vice-President for Information Infrastructure and
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EDUCATION

March 1990 PhD in Engineering, Kyoto University

March 1984 Master of Engineering, Kyoto University

March 1982 Bachelor of Engineering, Kyoto Institute of Technology

EXPERIENCES

Oct. 2022 – Present	Executive Vice-President for Information Infrastructure and Library Management
Apr. 2021 – Sept. 2024	Vice-President of Kyoto University Director General of Kyoto University IIMC
Oct. 2012 – Sept. 2022	Director General of Kyoto University Library Network Dean of Kyoto University Library
July. 2012 – Mar. 2024	Professor, Kyoto University, Japan Graduate School of Engineering, Department of Electrical Engineering Advanced Electrical System Theory Lab.
Aug. 2001 - June.2012	Professor, Kyoto University, Japan Graduate School of Engineering, Department of Electrical Engineering Power Conversion & System Control Lab.
Apr. 1997 - Jul. 2001	Associate Professor, Kyoto University, Japan Graduate School of Engineering, Department of Electrical Engineering Power System Lab.
Apr. 1994 - Mar. 1997	Associate Professor, Kansai University, Japan Department of Electrical Engineering
Apr. 1993 - Mar. 1994	Visiting Researcher, Cornell University, USA Sibley School of Aerospace and Mechanical Engineering

Apr. 1991 - Mar. 1993	Lecturer, Kansai University, Japan Department of Electrical Engineering
Apr. 1987 - Mar. 1991	Assistant Professor, Kansai University, Japan Department of Electrical Engineering
(Others)	
Apr. 2004 - Sept. 2004	part-time Lecturer, Nagoya Institute of Technologies, Japan Graduate School of Engineering
Apr. 2001 - Mar. 2004	part-time Lecturer, Tsukuba University, Japan Graduate School of Engineering
Apr. 1997 - Mar. 1999	part-time Lecturer, Kansai University, Japan Department of Electrical Engineering
Apr. 2000 - Sept. 2000	part-time Lecturer, Kyoto Institute of Technology, Japan School of Art & Engineering, Department of Electronic and Information Engineering
Apr. 1999 - Sept. 1999	same as above
Apr. 1996 - Sept. 1996	same as above

CURRENT RESEARCH TOPICS (Keywords)

1. Nonlinear Dynamics and Its Application
Analysis of nonlinear oscillations,
Controlling chaos,
Nonlinear coupled resonators,
Reservoir computing
2. Electric Power Network and System
Application of distributed power sources
Reachability analysis of nonlinear system
Power packetization and routing
3. Power Conversion and System Control
Applications of wide bandgap semiconductor power devices
Development of high frequency gate drives and its applications

(Keywords)

Nonlinear dynamics, Electric power network, System control, Data driven control

Activities

2016 /1 – 2019/12 Member Advisory Board, arXiv.org
2013 /5 - 2014/5 President of Engineering Sciences Society, the Institute of Electronics, Information and Communication Engineers (IEICE)
2012 /5 - 2013/5 Next-President of Engineering Sciences Society, the Institute of Electronics, Information and Communication Engineers (IEICE)
2010 /5 - 2012 /5 Vice President of Engineering Sciences Society, the Institute of Electronics, Information and Communication Engineers (IEICE)
2008 /5 - 2010 /5 ECC2009, International Program Committee, Member

2008 /5 - 2010 /5 ISCIE, Board Member (Editorial Affairs)
 2007 /5 - 2009 /5 IEEJ, Board Member (Globalization Promotion Charge)
 2006 /5 - 2008 /5 ISCIE, Board Member (Publication Affairs)
 2006 /5 - 2007 /5 IEICE, Technical Group of Nonlinear Problems, Chairman
 2005 /9 - 2006 /9 NOLTA2006, Steering Committee, Finance Chair
 2005 /5 - 2006 /5 IEICE, Technical Group of Nonlinear Problems, Vice-Chairman
 2004 /10 - 2006 /9 IEEJ, Research Group of Functional Application of Magnetic Suspension
 2004 /5 - 2006/4 IEEJ, IEEJ, Industrial Applications Society, Board Member (Editorial Affairs)
 2004 /3 - 2005 /9 International Symposium on Linear Drives for Industry Applications (LDIA) 2005,
 Program Committee, Chairman
 2004 /5 - 2004 /12 NOLTA04, Program Committee, Chairman
 2002 /4 - 2004 /3 IEEJ Research Group of Modeling and Simulation Technology on Power Electronics
 Circuits, Member
 2001 /4 - 2002 /4 Power Conversion Conference - Osaka 2002, Steering Committee, Program Committee,
 Member
 2001 /4 - 2002 /5 NOLTA'01, Technical Program Committee, Member
 2001 /4 - 2003 /3 IEEJ, Industrial Applications Society, Board Member
 2000 /6 - 2002 /5 IEEJ, Research Group of Coupling Problems on Magnetic Levitation Systems, Chairman
 2000 /4 - 2002/3 International Symposium on Linear Drives for Industry Applications (LDIA) 2001,
 Program Committee, Chairman
 1999 /7 - 2001 /8 IEEJ Research Group of Modeling and Simulation Technology on Power Electronics
 Circuits, Member
 1999 /4 - 2003 /3 IEEJ Annual Meeting, Japan Industry Applications Society Conference, Program
 Committee, Member
 1999 /4 - 2001 /3 IEEJ Research Group of Technical Terms on Linear Drive Systems, Member
 1998 /4 - 2006 /5 IEEJ, Technical Group on Linear Drive Technology, Member
 1998 /4 - 2001 /3 IEEJ, Research Group of Design Technology on Nonlinear Electronic Circuits, Member
 1998/4 - 2000 /4 IEEJ, Research Group of Nonlinear Technology in Magnetic Levitation Systems, Chairman
 1997/ 5 - 1998 /5 The Institute of Systems, Control, and Information Engineers (ISCIE), SCI'98 Steering
 Committee, Member
 1997 /2 - 1998 /4 International Symposium on Linear Drives for Industry Applications (LDIA) 1998,
 Program Committee, Member
 1996/ 5 - 2002 /4 IEICE, Technical Group of Nonlinear Problems, Member
 1996 /4 - 2001 /3 IEEJ, Research Group of Analytical Method for Nonlinear Circuit, Member
 1996 /4 - 1998/3 IEEJ Research Group of Application Technology of Magnetic Levitation System, Secretary
 1995 /5 - 1996 /4 IEICE, Workshop on Circuits and Systems in Karuizawa, Steering Committee
 Vice-Chairman
 1995 /10 - 1996 /10 International Symposium on Nonlinear Theory and Its Application '96, Technical
 Program Committee, Member
 1994 /7 - 2008 /5 IEICE, Nonlinear Theory and its Applications, Organizing Committee, Member
 1994 /7 - 1995 /12 International Symposium on Nonlinear Theory and Its Application (NOLTA) '95
 Steering Committee, Member

1994 /5 - 1996 /5 Institute of Electronics, Information and Communication Engineers (IEICE), Technical Group of Nonlinear Problems, Secretary

1994 /4 - 1996 /3 International Symposium of Linear Drive and Its Applications, Steering Committee, Member

1994 /4 - 1996 /3 IEEJ, Research Group of Industrial Application of Magnetic Levitation Technology, Assistant Secretary

1992 /4 - 1994/3 Institute of Electrical Engineers, Japan (IEEJ) , Research Group of Magnetic Levitation System, Assistant Secretary

1987 /4 - 2002/8 Japan Society for Power Electronics, Board member

Others:

Member of Council for Science and Technology of MEXT, Japan

Roles of Cabinet Office, JST

Editorial Activities

2026/3 – present, Editor of Nonlinear Dynamics, Springer-Nature

2018 /12 – 2025/1, IET, Editorial Board of The Journal of Engineering. Associate Editor

2015 /6 – 2019 /6 Editor-in-Chief, NOLTA, IEICE

2010 /9 – 2014/12 Journal of Vibration and Control, Editorial Board

2008 /5 - 2010/5 Journal of System, Control and Information, Vice-Editor

2008 /3 - 2010/12 European Journal of Control, Associate Editor

2006 /1 - 2008 /3 Journal of Circuits, Systems, and Computers, Associate Editor

2004 /5 - 2005 /10 IEICE, Transactions on Fundamentals of Electronics, Communications and Computer Science, NOLTA04 Special Section, Associate Editor

2003 /5 - 2004 /9 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Science on Fundamentals, NOLTA03 Special Section, Associate Editor

2002 /4 - 2003 /5 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Science, NOLTA02 Special Section, Associate Editor

2001 /4 - 2002 /5 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Science, Special Section on NOLTA01, Associate Editor

1999 /9 - 2000 /10 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Science, Special Section on NOLTA00, Secretary of Editorial Board, Associate Editor

1999 /4 - 2003 /3 IEEJ, Transactions on Industrial Applications, Associate Editor

1998 /4 – 2002 /5 IEICE, Transactions on Fundamentals of Electronics, Communications and Computer Science, Associate Editor

1997 /4 -1999 /3 The Institute of Systems, Control, and Information Engineers (ISCIE), Associate Editor

1995/10 - 1996 /10 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Science, Special Section on NOLTA96, Associate Editor

Invited International Talk

1. Phenomena and Boundary in Nonlinear Dynamical System, Plenary Talk, NOLTA2022(2022.12.14).
2. Tutorial Lecture, IEEEWiPDA-Asia2020 (2020.9.23)
3. Unite Nations Open Science Conference, United Nations Hamashold Library, NY, USA (2019.11.19)

4. Power packetization and routing, Conference on Intelligent Unmanned Swarm System Control Theory and Applications, Oct. 19, 2019, Beijing Institute of Technology, Beijing, China
5. Control of Intrinsic Localized Mode in Coupled Cantilever Array, IUTAM Symposium on Nonlinear Dynamics for Advanced Technologies and Engineering Design (NDATED), 27-30 July 2010, Aberdeen, UK.
6. Resonant manipulation of particles in nanosystems, RANM2009, Kuala Lumpur, Malaysia, Aug. 25 (2009).
7. Spectral Decomposition of Transient Vector Flow : Duffing System and Chaos, CCM8-ECCOMAS2008, Venice, Italy, June 30- July 5 (2008).
8. Dynamics in manipulation and actuation of nano-particles, ICAND2007, Kauai, Hawaii, USA, Sept. 24-27 (2007).
9. Wave's Travel by Instability of Mode in Coupled Pendulums, SIAM Conference on Applications of Dynamical Systems (DS07), May 28-June 1, 2007, Snowbird, Utah, USA (2007.5.31).
10. Control of nonlinear dynamics in nanosystem - from sensing to actuation -, Conference on Coupled Nonlinear Oscillators and Applications in Nanosystems, CNOAN, Santa Barbara, CA, USA (2007.5.25).
11. Georgia Tech., Lab. Seminar (non official; by Prof.K.Mischaikow), Nov. 9, 2005.
12. Georgia Tech., Lab. Seminar (non official; by Prof.R.Gregoriev), Nov. 7, 2005.
13. Princeton Univ., Dynamics Seminar (non official; by Prof. P.J.Holems), Dec. 13, 2002.

AWARDS

- | | | |
|-----|------|--|
| 1. | 2024 | IEICE, Achievement Award |
| 2. | 2021 | ISCIE, Best Paper Award 2021. |
| 3. | 2020 | Minister of Education, Culture, Sports, Science and Technology Award |
| 4. | 2019 | NOLTA Society: Distinguished Contribution Award. |
| 5. | 2019 | Best Paper Award 2019, SICE, #Takeda Award" |
| 6. | 2015 | ISCIE, Best Paper Award 2015 |
| 7. | 2015 | Best Paper Award 2015, IEICE, "Kiyasu Zenichi Prize" |
| 9. | 2015 | Best Paper Award 2015, MDPI energies |
| 10. | 2013 | ISCIE, Best Paper Award 2013 |
| 11. | 2011 | Fellow, IEICE |
| 12. | 2011 | Outstanding Contribution to Institute Award, IEICE-ESS |
| 13. | 2009 | ISCIE, Best Paper Award 2009 |
| 14. | 2008 | ACC2008, Best Presentation in Session Award |
| 15. | 2005 | IEEJ, IAS Activity Performance Award |
| 16. | 2001 | IEEJ, IAS Activity Performance Award |
| 17. | 1993 | IEEJ, Paper presentation award |
| 18. | 1990 | IEEJ Kansai chapter, Young engineers incentive award |

Institutes

IEEE Life Member, IEEJ, IEICE Fellow,
 APS(~2024), SIAM(~2024), ISCIE(~2025)

Publications

Books

1. Nonlinear Dynamics of Nanosystems (2009, Wiley-VCH Verlag), Eds. By Radons, G., Rumpf, Benno , Schuster, Heinz Georg, Chap. 9.
2. Handbook of Chaos Control (2007, Wiley-VCH), Eds. by Eckehard Schoel and Heinz Georg Schuster Chapter 36.
3. Power Electronics (in Japanese) (Asakura, 2000) Part.
4. Chaos Impact (Morikita, 1999) Translation.
5. Magnetic Levitation and Magnetic Bearing (in Japanese), pp.134-136 (Colona, 1993).

Published Paper (Reviewed)

1. Shiu Mochiyama, Kento Hiwatashi, and Takashi Hikihara, Upstream Allocation of Bidirectional Load Demand by Power Packetization, International Journal of Circuit Theory and Applications, Vol.53, Iss.12, December 2025, pp.6846-6855 (2025).
2. Rutvika Manohar and Takashi Hikihara, Design of a Stand-alone Hybrid Dispersed Generation Network Unified by Passivity Based Control, Royal Society Open Science, Vol. 11, Issue 7 (2024).
3. Yanran Wang, Tatsuya Baba, and Takashi Hikihara, Robot formation control in nonlinear manifold using Koopman operator theory, IEICE NOLTA, Vol.E15-N, No.2, pp.501-517 (2024).
4. Takahiro Mamiya, Shiu Mochiyama, and Takashi Hikihara, Router for wireless power packet transmission: Design and application to intersystem power management, IEEE Journal of Emerging and Selected Topics in Industrial Electronics, vol.5, no.2, pp.475-487 (2024).
5. Hirotaka Araki and Takashi Hikihara, Shift manipulation of intrinsic localized mode in ac driven Klein Gordon lattice, Physics Letters A, Vol. 493, 5, 129270 (2024).
6. Shiu Mochiyama and Takashi Hikihara, Stochastic Power Processing through Logic Operation of Power Packets, Royal Society Open Science, Vol.10, 230583 (2023).
7. Yuto Saito, Tomoya Shiono, Tomoyuki Tamura, Hiroshi Arai, Takashi Hikihara, and Nobuo Satoh, Power Distribution Technology based on Transmission of Power Packets via Conducting Metal Body, IEEJ Transactions on Industry Applications, Vol.144 No.1 pp.16–22 DOI: 10.1541/ieejias.144.16(2024).
8. Hajime Takayama, Suhei Fukunaga, and Takashi Hikihara: Digital-twin-compatible Optimization of Switching Characteristics for SiC MOSFETs using Genetic Algorithm, IEEE Journal of Emerging and Selected Topics in Industrial Electronics, Volume: 4, Issue: 4, 1024-1033 (2023)
9. Naoki Fujiwara, Nodoka Asayama, and Takashi Hikihara, Differential Game-based Safety Operation of Microgrids Including Virtual Synchronous Generator, Trans. of The Institute of System Control and Information Engineers, Vol.36, No.9, pp.316-326 (2023)
10. Shuhei Fukunaga, Hajime Takayama, and Takashi Hikihara, Slew rate control of switching transient for SiC MOSFET in boost converter using digital active gate driver, IET Power Electronics (2022)
11. Yohei Kono, Yoshihiko Susuki, and Takashi Hikihara, Characterizing scale dependence of effective diffusion driven by fluid flows, Phys. Rev. E 105, 045103 (2022).
12. Shiu Mochiyama, Kazuhiro Koto, and Takashi Hikihara, Routing optimization on power packet dispatching system based on energy loss minimization, IEICE NOLTA, Volume 13 Issue 1 Pages 139-148 (2022).

13. Hajime Takayama, Takafumi Okuda, and Takashi Hikiyara, Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior --- Preparation toward Universal Digitization of Power Switching, *International Journal of Circuit Theory and Applications*, 50, 183-196 (2022).
14. Nobuo Sato, Jimin Oh, and Takashi Hikiyara, Photo radiation pressure at resonance of frequency modulated micro cantilever, *NOLTA, IEICE*, Vol.E12-N, No.4, 718-725 (2021).
15. Shota Inagaki, Shiu Mochiyama, and Takashi Hikiyara, Electric Power Processing Using Logic Operation and Error Correction, "Royal Society Open Science, Vol. 8, Iss. 7, 8202344 (2021).
16. Yuma Murakawa and Takashi Hikiyara, Output Series-Parallel Connection of Passivity-Based Controlled DC-DC Converters: Generalization of Asymptotic Stability, *IEEE Trans. CAS I: Regular Papers*, Vol. 68, Iss. 4, 1750 - 1759 (2021).
17. Seongcheol Baek, Hiroyasu Ando, and Takashi Hikiyara, Decentralized Algorithms for Consensus-Based Power Packet Distribution, *NOLTA, IEICE*, Vol.12, No.2, pp.181 - 193 (2021).
18. Shiu Mochiyama, Takafumi Okuda, and Takashi Hikiyara, "Power Packet Dispatching with Shared Power Line: Experimental Verification for Industrial Applications", *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, Vol. 2, Iss. 2, 164-172 (2021)
19. Kazuki Hashimoto and Takashi Hikiyara, "Equivalent Circuit of Interleaved Air-Core Toroidal Transformer Derived from Analogy with Coupled Transmission Lines", *IEICE, ELEX*, 17 Issue 21 Pages 20200262 (2020).
20. Toko Mannari, Takafumi Okuda, and Takashi Hikiyara, "Prediction of restriction in output current by reactant flow in redox flow battery for compensating load variations", *International Journal of Circuit Theory and Applications*, Vol.49, Iss.4 1121-1132 (2021)
21. Yanran Wang and Takashi Hikiyara, "Two Dimensional Swarm Formation in Time-invariant External Potential: Modelling, Analysis, and Control", *AIP Chaos*, 30, 093145 (2020)
22. Manuel Sánchez Tejada, Takashi Hikiyara, "The Broken Symmetry of Output Currents in a Matrix Converter and Its Recovery Control", *Journal of Circuits, Systems, and Computers*, World Scientific, Vol.30, No.4, 2150072 (2020).
23. Shiu Mochiyama and Takashi Hikiyara, Impulsive torque control of biped gait with power packets, *Nonlinear Dynamics*, 102, pages 951–963 (2020).
24. Rutvika Manohar and Takashi Hikiyara, Phase synchronization of autonomous AC grid system with passivity-based control, *International Journal of Circuit Theory and Applications*, Vol. 48, Iss. 6, 906-918 (2020).
25. Shinji Katayama and Takashi Hikiyara, "Power packet router with power and signal switches for a single power packet", *IEEE Transactions on Circuits and Systems II: Express Briefs*, Vo.67, Iss.12, 3242-3246 (2020).
26. Kazuki Hashimoto, Takafumi Okuda, and Takashi Hikiyara, Measurement of internal magnetic flux density distribution in air-core toroidal transformer under high-frequency excitation, *Review of Scientific Instruments* 91, 044703 (2020).
27. Baek Seongcheol, Hiroyasu Ando, Takashi Hikiyara, Consensus-Based Distribution of Power Packets and Decentralized Control for Routing, *AIP, Chaos*, 30, 033115 (2020).
28. Rutvika Manohar and Takashi Hikiyara, Phase synchronization of autonomous AC grid system with passivity-based control, *International Journal of Circuit Theory and Applications*, Vol.48, Iss.6, 906-918 (2020).

29. Naomitsu Yoshida, Ryo Takahashi, and Takashi Hiki-hara, Power Regeneration from DC Motor with Bidirectional Router in Power Packet Dispatching System, IEEE Transactions on Circuits and Systems II: Express Briefs, Express Letter, Vol.67, Iss.12, 3088-3092 (2020).
30. Yuma Murakawa, Yohei Sadanda, and Takashi Hiki-hara, Parallelization of Boost and Buck Type DC-DC Converters by Individual Passivity-Based Control, IEICE Trans. A Vol.E103-A, No.3, 589-595 (2020).
31. Rutivika Manohar and Takashi Hiki-hara, Dynamic behaviour of a ring coupled boost converter system with passivity-based control, NOLTA, IEICE, Vol. 11, Iss. 1, 109-122 (2020).
32. Hikaru Hoshino, Yoshihiko Susuki, John Koo, and Takashi Hiki-hara. Structural Analysis and Control of a Model of Two-Site Electricity and Heat Supply. ASME. J. Dyn. Sys., Meas., Control. 2019;141(10):101004. doi:10.1115/1.4043703.
33. Fredrik Raak, Yoshihiko Susuki, Shinya Eguchi, and Takashi Hiki-hara, Role of deterministic electromechanical conversion in short-term wind power fluctuations: A case study in Japan, Physical Review Applied, Phys. Rev. Applied 12, 034013 (2019).
34. Toko Mannari; Takafumi Okuda, Takashi Hiki-hara, Model for charging/discharging dynamics of cells in redox flow battery with transport delay, Pyisica Scripta Volume 94, 1-9 (2019).
35. Sinji Katayama, Ryo Takashashi, and Takashi Hiki-hara, Experimental Verification of Distributed Energy on Demand Control Using Wireless Communication on Power Packet Dispatching System, Trans. IEICE Vol. J102-A No. 9 pp. 240-248 (2019).
36. Nodoka Asayama, Takashi Hiki-hara, On the Use of Reachable Set for Safe Operation in Microgrid, Trans. of the Institute of Systems, Control and Information Engineers, Vol.32, Iss.4, 137-144 (2019)
37. Shiu Mochiyama and Takashi Hiki-hara, Packet-based feedback control of electrical drive and its application to trajectory tracking of manipulator, International Journal of Circuit Theory and Applications, Vol.47, Iss. 4, 612-632 (2019). <http://dx.doi.org/10.1002/cta.2603>
38. Manuel Sanchez and Takashi Hiki-hara, Symmetry Recovering of an AC/AC Converter Working in a Chaotic Regime, NOLTA, IEICE, Vol. 10, Issue 2, pp.131-139 (2019).
39. Manuel Sanchez, Takafumi Okuda, and Takashi Hiki-hara, "Stabilization of Mode in Imbalanced Operation of Matrix Converter by Time-Delayed Feedback Control" , International Journal of Circuit Theory and Applications, John Wiley, DOI: 10.1002/cta.2556.
40. Shinya Nawata, Atsuto Maki, Takashi Hiki-hara, Power packet transferability via symbol propagation matrix, Proc. R. Soc. A 2018 474 20170552; DOI: 10.1098/rspa.2017.0552.
41. Shinya Nawata, Ryo Takahashi, and Takashi Hiki-hara, "Design possibility for covert power packet transfer using asynchronous sampling", NOLTA, IEICE, Vol. 9 Issue 3, pp. 374-381 (2018). <https://doi.org/10.1587/nolta.9.374>
42. Y. Kono, Y. Susuki, H.Mitsunori I. Mezic, and T.Hiki-hara, Multiscale Modeling of In-Room Temperature Distribution with Human Occupancy Data: A Practical Case Study, Journal of Building Performance Simulation, 11, 2, 145-163 (2018).
43. M.Shintani, Y. Nakamura, K.Oishi, M. Hiromoto, T.Hiki-hara, and T.Sato, "Surface-Potential-Based Silicon Carbide Power MOSFET Model for Circuit Simulation", IEEE Trans. Power Electronics (2018.2. 13). DOI: 10.1109/TPEL.2018.2805808
44. Takafumi Okuda, Takashi Hiki-hara, Skew Calibration with High-Power and High-Speed Pulse Generator for Oscilloscope, IEEJ Trans. on Electronics, Information and Systems, Vol.138, Iss.1, 37-40 (2018).

45. Ryosuke Meda, Takafumi Okuda, Takashi Hikihiro, Analysis of dynamic characteristics of SiC SBDs at high switching frequency based on junction capacitance, *Japanese Journal of Applied Physics* 57(4S):04FF01 DOI10.7567/JJAP.57.04FF01
46. Rochelle Nieuwenhuis, Madoka Kubota, M. R. Flynn, Masayuki Kimura, Takashi Hikihiro, and Vakhtang Putkaradze, Dynamics regularization with tree-like structures, *Applied Mathematical Modelling*, Volume 55, Pages 205-223 (2018).
47. Toko Mannari and Takashi Hikihiro, Transient Behavior of Redox Flow Battery Connected to Circuit Based on Global Phase Structure, *NOLTA, IEICE*, Volume 9, Issue 1, 137-147 (2018) .
48. Fredrik Raak, Yoshihiko Susuki, Kazuhisa Tsuboki, Masaya Kato, and Takashi Hikihiro, Quantifying Smoothing Effects of Wind Power via Koopman Mode Decomposition: A Numerical Test with Wind Speed Predictions in Japan, *NOLTA, IEICE*, Vol.8, Iss.4, 342-357(2017).
49. Shiu Mochiyama, Ryo Takahashi, and Takashi Hikihiro, Close-loop Angle Control of Stepper Motor Motor Fed by Power Packets, *IEICE Trans. A*, Vol.E100-A, No.7, pp.1571-1574, (2017).
50. Shiu Mochiyama, Ryo Takahashi, and Takashi Hikihiro, Trajectory control of manipulator fed by power packets, *International Journal of Circuit Theory and Applications*, vol.45, no.6, pp.832--842, (2017).
51. Tsukasa Saito, Yoshihiko Susuki, and Takashi Hikihiro, Mode Switching Control for Output Levelling of In-Home Electric Power System : Experiment and Performance Evaluation, *Trans. IEICE*, J100-A(5), 183-194 (2017).
52. H.Hoshino, Y.Susuki, Koo T.John, and T.Hikihiro, Nonlinear Control of Combined Heat and Power Plants in a Two-site Regional Energy System -Simultaneous Regulation of Electricity and Gas Flow, *Transactions of the Institute of Systems, Control and Information Engineers*, Vol. 30 Iss. 5, 157-166 (2017).
53. Yohei Kono, Yoshihiko Susuki, and Takashi Hikihiro, Applications of Koopman Mode Decomposition to Modeling of Heat Transfer Dynamics in Building Atriums-II, *Transactions of the Society of Instrument and Control Engineers*, Vol.53, No.2, 188--197 (2017).
54. Yohei Kono, Yoshihiko Susuki, Mitsunori Hayashida, Takashi Hikihiro, Applications of Koopman Mode Decomposition to Modeling of Heat Transfer Dynamics in Building Atriums-I—Effective Heat Diffusion by Small-scale Air Movement—, *Transactions of the Society of Instrument and Control Engineers*, Vol.53, No.2, 123--133 (2017).
55. Michihiro Shintani, Yohei Nakamura, Masayuki Hiromoto, Takashi Hikihiro, and Takashi Sato: "Measurement and Modeling on Gate-Drain Capacitance of Silicon Carbide Vertical MOSFET," *Japanese Journal of Applied Physics*, Vol.56, No.4S, 04CR07.
56. Ryo Takahashi, Shun-ichi Azuma, Mikio Hasegawa, Hiroyasu Ando, and Takashi Hikihiro, Power Processing for Advanced Power Distribution and Control, *IEICE Transaction B Position Paper*, Vol. E100.B, Iss.6, 941-946 (2017).
57. Atsushi Yao and Takashi Hikihiro, Reprogrammable logic-memory device of a mechanical resonator, *International Journal of Non-linear Mechanics*, Vol.94, 406-416 (2017).
58. Masayuki Kimura, Yasuo Matsushita, Takashi Hikihiro, Parametric resonance of intrinsic localized modes in coupled cantilever arrays, *Physics Letters A*, 380, 2823–2827(2016).
59. Shinya Nawata, Ryo Takahashi, and Takashi Hikihiro, "Up-stream Dispatching of Power by Density of Power Packet" *IEICE Trans. Vol.E99-A*, No.12, pp.2581-2584 (2016).
60. Ryo Takahashi, Shunichi Azuma, and Takashi Hikihiro, Power Regulation with Predictive Dynamic Quantizer in Power Packet Dispatching System, *IEEE Transactions on Industrial Electronics*, 63(12),

7653-7661 (2016).

61. Yoshihiko Susuki, Igor Mezic, Fredric. Raak, and Takashi Hikihara, Applied Koopman operator theory for power systems technology, *Nonlinear Theory and Its Applications*, IEICE, Vol. 7(4), 430-459 (2016).
62. Edmon Perkins, Masayuki Kimura, Takashi Hikihara, and Balakumar Balachandran, Effects of Noise on Symmetric Intrinsic Localized Modes, *Nonlinear Dynamics*, 85(1) (2016).
63. Yanzi Zhou, Ryo Takahashi, and Takashi Hikihara, Power Packet Dispatching with Features on Safety, *NOLTA*, IEICE, Vol.7, No.2, 250-265 (2016).
64. H. Hoshino, Y. Susuki, and T. Hikihara, A Lumped-Parameter Model of Multiscale Dynamics in Steam Supply Systems, *ASME Journal of Computational and Nonlinear Dynamics* 11(6), 061018 (2016).
65. Fredric Raak, Yoshihiko Susuki, and Takashi Hikihara, Data-driven partitioning of power networks via Koopman mode analysis, *IEEE Transactions on Power Systems*. Vol.31, Iss.4 (2016).
66. Y.Zhou, R.Takahashi, N.Fujii, and T.Hikihara, Power Packet Dispatching with Second-Order Clock Synchronization, *International Journal of Circuit Theory and Applications*, published(online(DOI)).
67. M.Kubota, R.Takahashi, and T.Hikihara, Active and Reactive Power in Stochastic Resonance for Energy Harvesting. *IEICE Transactions* 98-A(7): 1537-1539 (2015).
68. K.Nagaoka, K.Chikamatsu, A.Yamaguchi, K.Nakahara, and T.Hikihara, High-speed gate drive circuit for SiC MOSFET by GaN HEMT, *IEICE Electronics Express* Vol. 12, No. 11 pp. 20150285 (2015).
69. A.Kordonis , R.Takahashi, D.Nishihara, and T.Hikihara, Three-Phase Power Router and its Operation with Matrix Converter toward Smart-Grid Applications, *Energies*, 8, 3034-3046 (2015).
70. A.Kordonis and T.Hikihara, "Dynamic model of direct matrix converter and its experimental validation", *Int. J. Circ. Theor. Appl.*, Published online in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/cta.2072 (2015).
71. R.Takahashi, K.Tashiro, and T.Hikihara, Router for Power Packet Distribution Network: Design and Experimental Verification, *IEEE Trans. on Smart Grid*, Vol.6, No.2, 618-626 (2015)..
72. Y-Z.Zhou, R.Takahashi, and T.Hikihara, Realization of Autonomous Clock Synchronization for Power Packet Dispatching, *IEICE Trans. A*, Vol.E98-A,No.2, (2015) .
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Conference Paper: more than 200,

Patents: 13

Evaluation:

(Google Scholar) Citations 5584, h-index 39, i10-index 116

(WoS) Times Cited 2799, Publications in WoS 224, h-index 28

(Elsevier, Updated science-wide author databases of standardized citation indicators)

Top 2% nominated (career), rank 518, 379, h-index 25, hm24 14.4524

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Top percentage rank 0.44% (All field), 0.49% (Engineering), 0.24% (Electric power)

0.92% (Nonlinear system)

<https://scholargps.com/scholars/66955271102488/takashi-hikihara>

List of PhD Students and Title of Dissertations

Yuma Murakawa (2023.3.24)

Application of Passivity-Based Control to Series-Parallel Connected DC-DC Converters and their Circuit Characteristics

Shinji Katayama (2022.3.24)

A Feasible Design of Power Packet Dispatching System

Yohei Kono (2021.3.23)

Diffusion and Drift Transport between Zones in Building Temperature Fields and its Data-Driven Modeling (in Japanese)

Baek Seongcheol (2021.3.23)

Dynamical Analysis and Decentralized Control of Power Packet Network

Kazuki Hashimoto (2021.3.23)

Analysis and Design of Air-Core Transformer Based on Internal Magnetic Flux Density Distribution for High-Frequency Power Converter

Rutvika Manohar Nandan (2021.3.23)

Design of Distributed Stand-alone Power Systems using Passivity-Based Control(

Toko Mannari (2020.11.23)

Mass Transport and Discharging Dynamics of Redox Flow Battery for Power Supply

Manuel Antonio Sanchez Tejada (2019.9.24)

Appearance of Symmetry Breaking in AC/AC Converters and Its Recovery Methods

Shiu Mochiyama (2019.9.24)

Power-packet Based Control and Its Application in Distributed System

Fredrik Raak (2017.9.25)

Data-driven analysis of wind power and power system dynamics via Koopmanmode decomposition

Masatsugu Nakano (2017.9.25)

Research on Loss Analysis and Torque Ripple Reduction for High Performance of Permanent Magnet Synchronous Motors (in Japanese)

Hikaru Hoshino (2017.3.23)

Analysis and Control of Multiscale Dynamics in Regional Electricity and Heat Supply Systems

Shinya Nawata (2017.3.23)

Design of Electrical Energy Network Based on Power Packetization

Yanzi Zhou (2015.9.24)

Power Packet Dispatching Based on Synchronization with Features on Safety

Madoka Kubota (2015.3.23)

Energy Harvesting Characteristics of Nonlinear Oscillators under Excitation

Atsushi Yao (2015.3.23)

Logic and memory devices of nonlinear microelectromechanical resonator

Alexandros Kordonis (2015.3.23)

DYNAMIC ANALYSIS AND MODELING OF AC/AC POWER CONVERTERS FOR APPLICATIONS TO SMART-GRID SOLUTIONS

Masataka Minami (2013.3.25)

Research on Control Based on Passivity and Phase Synchronization Method in Grid Connection of Distributed Power Sources (in Japanese)

Tsuguhiro Takuno (2012.3.23)

High Frequency Switching of SiC Transistors and its Application to In-home Power Distribution

Suketu D. Naik (2011.8.26)

Investigation of Synchronization in a Ring of Coupled MEMS Resonators

Yuuichi Yokoi (2011.3.23)

Energy Conversion and its Control by Synchronization of Pendulum

Phankong Nathabhat (2010.9.24)

Characterization of SiC Power Transistors for Power Conversion Circuits Based on C-V Measurement

Masayuki Kimura (2009.3.23)

Studies on the Manipulation of Intrinsic Localized Modes in Coupled Cantilever Arrays

Mei Hua Li (2009.3.23)

Fundamental Research on the Dynamic Model Based on the Chemical Reactions of Redox Flow Batteries (in Japanese)

Kohei Yamasue (2007.3.23)

Studies on Time-Delayed Feedback Control of Chaos and its Application to Dynamic Force Microscopy

Katsunori Asano (2006.3.23)

Fundamental Research on the Application of High Voltage Resistant SiC FET and Diodes to Power Conversion Circuits (in Japanese).

Yoshihiko Susuki (2005.3.23)

Transient Dynamics and Stability Boundaries in Electric Power System with DC Transmission

List of Funding

1. Grant-in-Aid for Scientific Research (Main proposal)

1991.4-1992.3

Grant-in-Aid for Encouragement of Scientists(A) 03750201 Amount no record

Development of absorbtio type magnetic levitation system by linear induction motor

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Study on measurement of traveking magnetic field vector of linear otor by magnetic birefringence

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Grant-in-Aid for Scientific Research (C) 10650276, ¥ 3,500 K

Restriction of nonlinear spatio-temporal propagation of vibration in magnetically suspended elastica by power electronics technology

2000.4 - 2002.3

Grant-in-Aid for Scientific Research (C) 12650276, ¥3,500K

Study on method for restriction of electric-mechanic coupled vibration in magnetically suspended system by power conversion and control

2002.4 - 2004.3

Grant-in-Aid for Scientific Research (C) 14550264, ¥2,900K

Development of simulator of power system with DC power transmission for verification method of stability control

2006.4 - 2008.3

Grant-in-Aid for Scientific Research (B) 18360137, ¥18,590K

Design and Integration of power distribution circuit with SiC power devices with EMC estimation

2009.4 - 2011.3

Grant-in-Aid for challenging Exploratory Research 21656074, ¥3,380K

Development of mechanism for operating nano/micro structure and scavenging micro-energy

2014.4 - 2017.3

Grant-in-Aid for challenging Exploratory Research 26630176, ¥ 3,770K

Development of nonlinear resonator and converter circuit for energy harvesting from wide range spectrum of vibration

2020.4 – 2023.3

Grant-in-Aid for Scientific Research (B) 20H0215, ¥17,810K

Fundamental study of basic technology to realize digitization of small-capacity power transmission

2023.4 – 2026.3

Grant-in-Aid for Scientific Research (B) 20H01399, ¥8,970K(2023)

Development of fast self regulating digital active gate driver for SiC MOSFET

2. Grant-in-Aid for Scientific Research (assisted proposal)

1992.4 - 1993.3

Grant-in-Aid for Scientific Research (C) 04650285

Study on characteristics of superconducting motor with high Tc superconductor

1997.4 - 1999.3

Grant-in-Aid for Scientific Research (C) 09650441

Fundamental research on transient behavior of power system and evaluation of instability criteria

2000.4 - 2002.3

Grant-in-Aid for Scientific Research (C) 12834006

Reviewing study on power and information transfer via synchronization in coupled electrical circuit

2000.4 - 2001.3

Grant-in-Aid for Scientific Research (C)(1) 12895006

Novel development of real-time control -

Systematic approach to theory of hybrid dynamical system

2004.4 - 2006.3

Grant-in-Aid for Exploratory Research 16656089

Development of hybrid simulator for next generation power system analysis

3. Governmental funding

2008.10 - 2013.3

Knowledge Clusters Project (MEXT) ¥69,800K

2009.9 - 2014.3

NICT Project #142 ¥85,000K

Development on information and energy network technology for integration

2011.10 - 2012.9

Advance Low Carbon Technology (ALCA) (JST) ¥4,680K

Network Design of novel power transfer by power processing

2013.12 - 2018.3

Kyoto Area Super Cluster Program (JST) ¥42,700K

Construction of Highly-efficient Energy Utilization System for Realizing Clean and Low-environmental Load Society

2014.11 - 2016.3

SIP Program (Total ¥193,899K)

Development of low power and small sized concentrated circuits and power processing technology by SiC power devices

2018.10 - 2022.3

SIP Program, ¥8,679K

Development of ultra-fast digital controller for WBG semiconductor

4. Private Funding

2002.4 - 2003.3

Iwatani Naoji Foundation ¥2,000K

Development of linkage method of distributed power sources to power system and study on its stability

2004.4 - 2005.3

JFE 21 Century Foundation ¥2,000K

Fundamental study on design of energy network by passive micro grid by distributed power sources

2008.4 - 2010.3

CASIO SCIENCE PROMOTION FOUNDATION ¥3,000K

Development of gate drive circuit of SiC power devices for high speed switching